

Wuxi YuHuang Transmission Technology Co., Ltd

无锡羽凰传动科技有限公司



YH8-520P

[Linear Actuator]

YH-520P Selection Guide

Product Overview

1. DC brushed motor
2. Stroke: **30 mm to 1000 mm** (customizable)
3. Speed: **7–50 mm/s**, optional
4. Maximum load capacity: **16,000 N**
5. Operating temperature: **–40 °C to +65 °C**
6. Protection rating: **IP66**



Understanding the Linear Actuator

S: Stroke

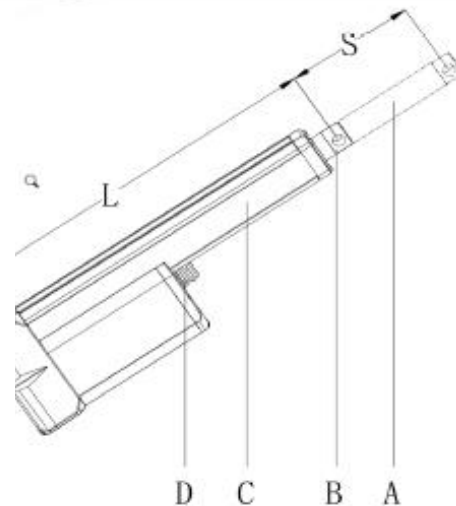
L: Installation distance

A: Stainless steel inner tube

B: Zinc-plated threaded connector

C: Aluminium alloy housing

D: Waterproof cable connector



Stroke	The maximum travel distance of the actuator, i.e., the dimensional change between fully retracted and fully extended positions. <i>(Customizable)</i>
Installation Distance	The overall length of the actuator in the fully retracted state. <i>(Customizable)</i>
Front-End Connector	Multiple option available.
Rear-End Connector	Multiple option available.
Mounting Angle	Optional: 0° / 30° / 60° / 90°
Dynamic Load	The maximum load the actuator can withstand while in motion.
Self-locking Force	The maximum load the actuator can withstand when stationary.
Protection Rating	IPXX: The first digit indicates dust protection; the second digit indicates water protection. Higher numbers indicate better protection performance.
Duty Cycle	Continuous working time / rest time, expressed as a percentage (a / (a + b)).
Speed	Output speed under rated load conditions.
Hall Sensor	Outputs voltage pulse signals. Stroke displacement is calculated by counting pulses, and motion direction can be identified based on waveform phase.

Model Selection Code	520P	X	X	X	X	X	X	X	X	X
Actuator Model										
YH-520P Model Code: 520P										
Voltage										
12V		1								
24V		2								
36V		3								
48V		4								
Load and Speed										
Refer to Table 1.0										
Travel										
Refer to Table 4.0										
Installation Distance										
Refer to Table 4.0										
Front End Connector Type										
Ø14mm hole							1			
Ø16mm hole							2			
Ø20mm hole							3			
Ø14mm hole with 12mm U-shaped groove							4			
Ø16mm hole with 12mm U-shaped groove							5			
Ø20mm hole with 12mm U-shaped groove							6			
M16x2.0x20L external thread							7			
M16x2.0x20L internal thread							8			
Ø16 rod-end spherical bearing							9			
Rear End Connector Type										
Ø14mm hole							1			
Ø16mm hole							2			
Ø20mm hole							3			
Ø14mm hole with 10mm U-shaped groove							4			
Ø16mm hole with 10mm U-shaped groove							5			
Ø20mm hole with 10mm U-shaped groove							6			
M16x2.0x20L external thread							7			
M16x2.0x20L internal thread							8			
Ø16 rod-end spherical bearing							9			
Power cord										
1.0m power cord								1		
1.5m power cord								2		
2.0m Power Cord								3		
Customizable								4		
Ingress Protection Rating										
IP65 (Standard)										
Signal Output										
No Signal Output									0	
External Magnetic Limit Switch Signal									1	
Hall Signal									2	
Potentiometer signal output									3	
Screw Type										
Ball Screw										1

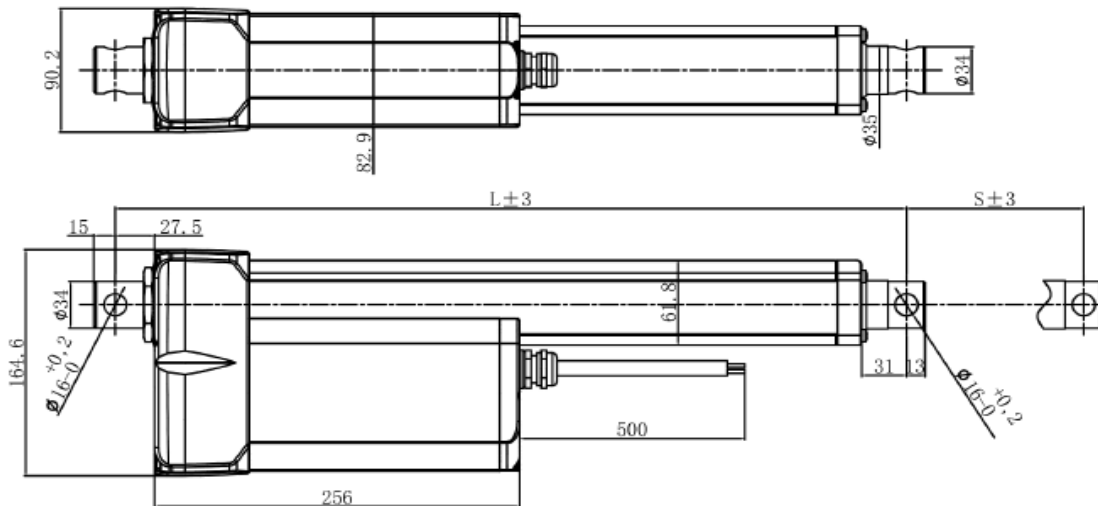
Table 1.0: Load and Speed Reference

Transmission Code	Maximum Dynamic Load	Self-Locking Force	Reduction Ratio	Pitch	Speed $\pm 10\%$		Maximum Stroke
					No Load	Full Load	
	(N)	(N)	-	(mm)	(mm/s)	(mm/s)	(mm)
A	16000	20000	45:1	5	7.5	5.5	1500
B	8000	12000	45:1	10	15	11.0	1500
C	4000	6000	45:1	20	30	23.0	1500
D	12000	15000	35:1	5	9.4	7.0	1500
E	6000	8000	35:1	10	18	14.0	1500
F	3000	5000	35:1	20	35	26.0	1500
G	9000	12000	25:1	5	13	10.0	1500
H	4500	7000	25:1	10	26	20.0	1500
I	2000	3000	25:1	20	50	40.0	1500

1. All data measurements are conducted at an ambient temperature of 20 °C with a stable power supply voltage.

2. The *customizable maximum stroke* is influenced by multiple factors, including load, speed, direction of force, and duty cycle. Therefore, it must be evaluated based on the actual application conditions. If your required parameters exceed the available range, please contact us for further assessment.

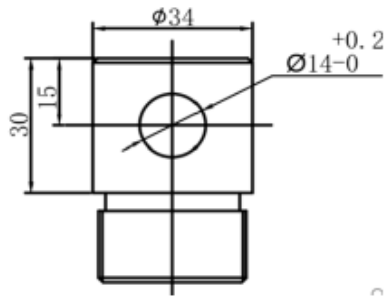
Table 4.0: Reference for Stroke Length and Installation Distance



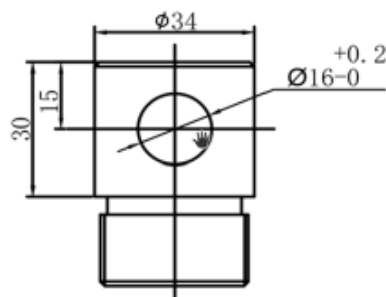
Note:

- When $30 \leq \text{Stroke} \leq 120\text{mm}$, $L = 330 \text{ mm}$ When $120 < \text{Stroke} \leq 600\text{mm}$, $L = S + 215$
When $600 < \text{Stroke} \leq 1000\text{mm}$, $L = S + 300$ When $1000 < \text{Stroke} \leq 1500\text{mm}$, $L = S + 400$
- The hole orientation shown above is **90°**.
- When the front end connector is a **rod-end joint bearing** or a **customized connector**, the above installation dimension rules do not apply.

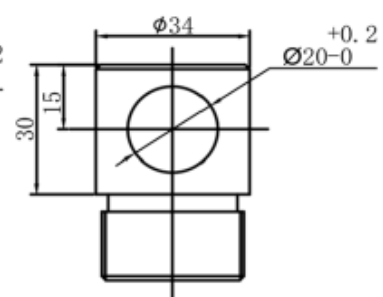
Front-End Connector



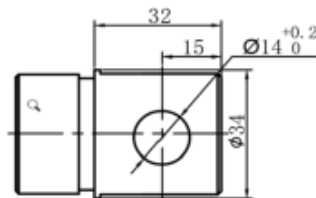
01



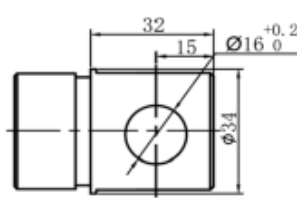
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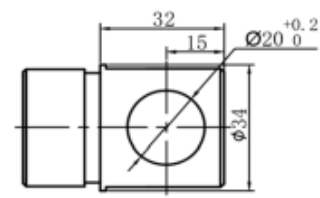
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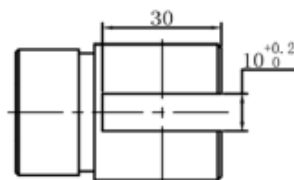
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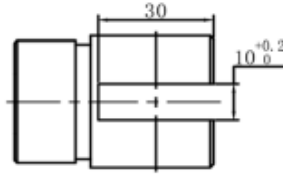
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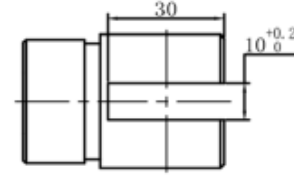
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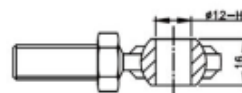
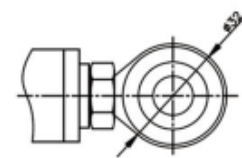
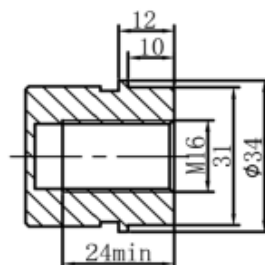
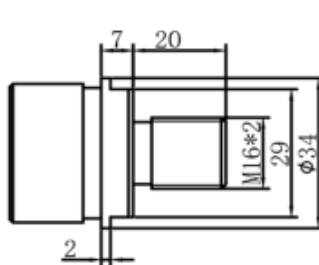
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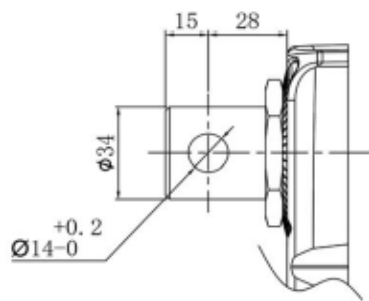
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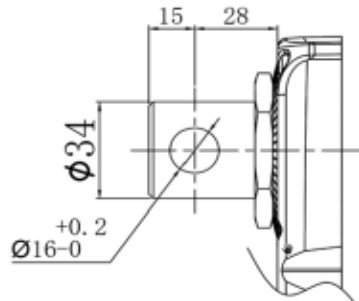
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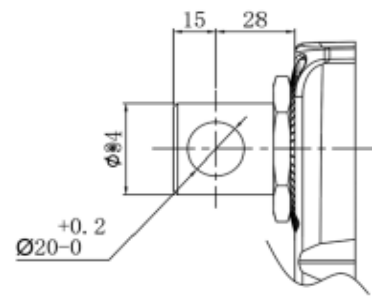
Rear-End Connector



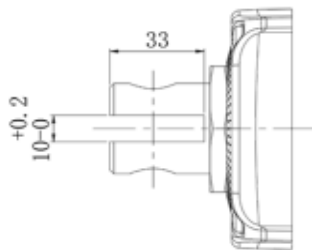
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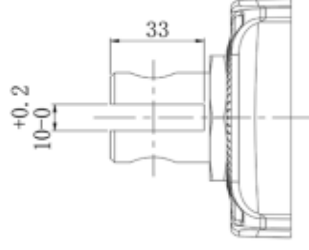
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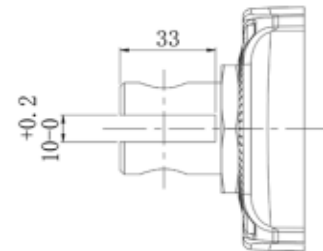
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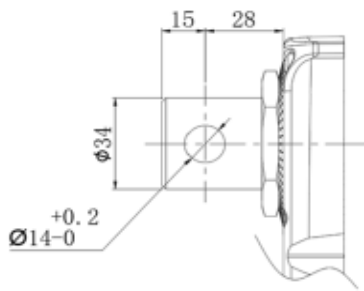
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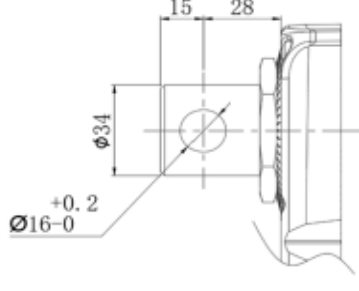
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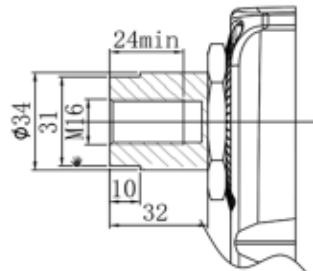
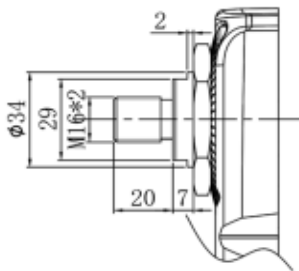
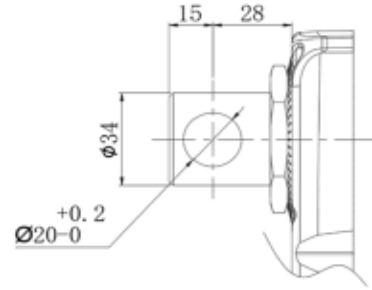
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07



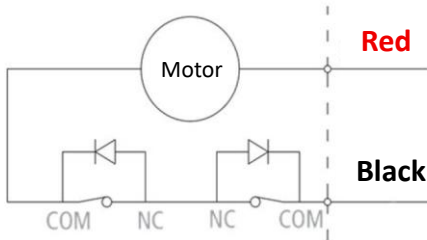
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Signal output

0. Built-in limit switch, no signal feedback.

The actuator is equipped with upper and lower limit switches as standard. When the actuator reaches full extension or full retraction, the limit switches automatically cut off the power.

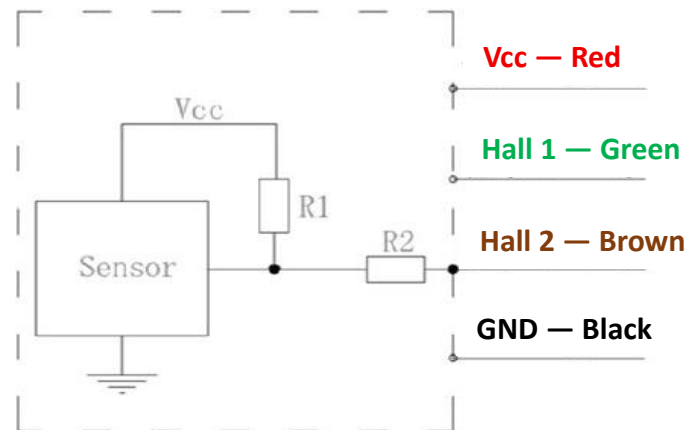


Extension / Retraction Wiring Configuration		
Wire Colour	Black	Red
Extend	-	+
Retract	+	-

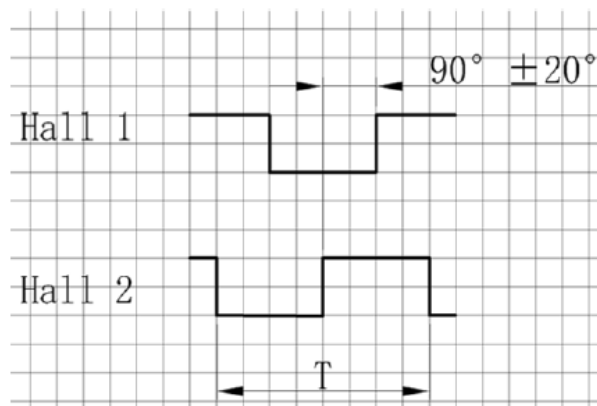
1. External magnetic limit switch signal

2. Hall sensor signal feedback

Transmission Code	Feedback Method	Number of Magnetic Poles	Pulse Resolution (pulse/mm)
A	Hall signal feedback	4 poles	35.68
B			17.84
C			8.92
D			28.28
E			14.14
F			7.07
G			20.06
H			10.03
I			5.02



Operating voltage range: $V = 5 \sim 15 \text{ V}$



Pulses per 1mm per single Hall sensor

3. Potentiometer signal feedback

Transmission Code	Maximum Supported Stroke (mm)	Resistance per mm (Ω/mm)
A, D, G	610	16.393
B, E	1220	8.197
C, F, I	2440	4.098

